25900294 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACCHE200 - FORENSIC CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries **1** marks

1. Naloxone is used as an antidote for poisoning by [K] / [CO1]

   a). Organophosphates b). Cyanide

   c). Opioids d). Iron

2. Which device is commonly used to detect metallic objects during security checks? [K] / [CO2]

   a). Ultrasound device b). Metal detector

   c). Infrared Scanner d). X- Ray scanner

3. What is the primary function of a police dog in crime scene investigation [K] / [CO4]

   a). Identify paint types b). Detect explosives

   c). Track scent trails d). Analyze footprints

4. Human hair is considered a reliable biomarker of arsenic exposure because [U] / [CO1]

a). It contains melanin

b). It incorporates arsenic from the bloodstream during growth

c). It resists environmental contamination

d). It is rich in proteins

5. What is the principle involved in detecting the presence metals using metal detector devices? [K] / [CO2]

a). Generation of electric field

b). Doppler Effect

c). Generation of electromagnetic field

d). Piezoelectric effect

6. The typical yellow color of urine is primarily due to which of the following? [K] / [CO4]

a). Hemoglobin

b). Urea

c). Urochrome

d). Uric acid

7. Which of the following is an example of a technical document? [K] / [CO3]

a). User guide for software

b). Personal diary

c). Legal affidavit

d). Cover letter

8. Cadmium accumulation in fish particularly affects [K] / [CO1]

a). Muscles and heart

b). Brain and lungs

c). Kidneys and liver

d). Skin and bones

9. Which explosive is used for blasting in quarries? [K] / [CO2]

a). TNT

b). Gelatin Stick

c). Black Powder

d). PETN

[1x9 = 9]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **4** marks

10. State four signs and symptoms of cyanide poisoning. [K] / [CO1]

11. Differentiate between toxic and corrosive chemicals? [U] / [CO2]

12. Explain the main security features in Indian coins [K] / [CO3]

13. How can DNA fingerprinting be employed in transplantations? [U] / [CO4]

14. List four postmortem findings in strychnine poisoning. [K] / [CO1]

15. Define an incendiary bomb and state its primary purpose. [K] / [CO2]

16. What are the physical side effects of long-term anabolic steroid use? Mention at least four. [K] / [CO4]

[4x5 = 20]

Part C

Short Essay Questions

Answer any **3** questions

Each question carries **7** marks

17. Discuss the classification of poisons according to their origin with examples [U] / [CO1]

18. Describe with a diagram the main parts of a bullet and their composition. [U] / [CO2]

19. Describe the principle and different steps of using a diamond tester for diamond authentication. [U] / [CO3]

20. Explain the application of ultraviolet rays in forensic document examination with at least four examples. [U] / [CO3]

21. Explain different types of fibre analysis conducted in forensic labs [U] / [CO4]

[7x3 = 21]